

RB-HDRDiff: Reducing Banding Artifacts for Temporally Consistent SDR-to-HDRTV Reconstruction via Diffusion Models

Anonymous Submission

No Institute Given

Abstract. Learning-based SDRTV-to-HDRTV reconstruction methods often suffer from severe banding artifacts due to the aggressive compression of visual information into low-dimensional latent spaces. Besides, most existing approaches provide limited support for maintaining temporal coherency when applied to videos, while temporally consistent HDRTV datasets remain scarce. To address these challenges, we propose RB-HDRDiff, a diffusion-based HDRTV reconstruction framework that utilizes the non-compressive latent space of diffusion models to reduce banding artifacts and improve visual fidelity. RB-HDRDiff adopts a two-stage training strategy. First, a diffusion model is pre-trained on high-quality HDR images to learn a robust HDR prior. Second, we introduce HDR Temporal Consistency Refiner, a video refinement module that improves inter-frame consistency by incorporating information from neighboring frames while maintaining computational efficiency. We further introduce a new HDR video dataset for temporal consistent HDRTV reconstruction. Extensive experiments demonstrate that RB-HDRDiff outperforms state-of-the-art methods in terms of visual quality, banding artifact reduction, and temporal stability.

Keywords: SDR-to-HDR conversion · Diffusion model · Video enhancement

1 Introduction

High Dynamic Range Television (HDRTV) delivers substantially enhanced visual quality compared with conventional Standard Dynamic Range Television (SDRTV), offering superior highlight and shadow reproduction [32], an expanded color gamut [5], and overall imagery closer to human perception. With the rapid adoption of HDR-capable displays and standards such as BT.2020, BT.2100, and BT.2446 [9, 10, 12], the demand for high-quality HDRTV content continues to increase. However, most legacy broadcasting and online media remain in SDR format due to the high cost of HDR production equipment [5, 32] and the complexity of HDR workflows [5]. Consequently, converting existing SDRTV footage into HDRTV-compliant content, also referred to as inverse tone mapping (ITM) [8], has become crucial for streaming platforms, broadcasters, and display manufacturers.



Fig. 1: Visual comparison of HDRTV reconstruction results. Our method effectively reduce banding artifacts in the sky region (red boxes), achieving smoother luminance transitions and closer consistency with the ground truth.

Recent advancements in deep learning-based ITM method [1, 32, 8, 5, 14, 13, 23, 29] have demonstrated strong quantitative performance. Despite promising progress, their visual results often suffer from banding artifacts and incomplete detail recovery. This limitation arises from their use of encoder-decoder architectures that compress SDRTV signals into low-dimensional latent spaces. Such aggressive feature compression results in the loss of fine-grained information, particularly in highlight regions, making it challenging to accurately reconstruct complex highlight structures. Consequently, banding artifacts and incomplete detail recovery are prevalent in these regions [38, 6], as illustrated in Figure 2. These artifacts not only degrade the perceived HDRTV quality but also limit the applicability of existing methods in high-end production scenarios. Moreover, existing ITM methods are predominantly trained on static images, with limited consideration for the characteristics of video data. Temporal coherence is therefore not sufficiently considered during training, leading to noticeable temporal inconsistencies when these models are directly applied to SDRTV-to-HDRTV reconstruction.

To address these challenges, we propose RB-HDRDiff, a diffusion-based HDRTV reconstruction framework designed to reduce banding artifacts while preserving high-quality HDR details. Diffusion models operate in a non-compressive, high-dimensional latent space that is perceptually aligned with natural image statistics [21], making them inherently suitable for mitigating the information loss commonly encountered in encoder-decoder based ITM pipelines. Our framework adopts a two-stage training strategy. In the first stage, a diffusion model is pre-trained on high-quality HDR images to learn a strong HDR prior and its parameters are frozen thereafter. In the second stage, we introduce an HDR Temporal Consistency Refiner to improve temporal coherence in video frames. The refiner consists of three HDR Frame Refine Units (HFRU), which process the neighboring frames at time steps $t - 1$, t , and $t + 1$, respectively. The outputs of these units are concatenated into a temporal attention layer to enhance temporal consistency and preserve fine details in the HDR reconstruction. To facilitate research on temporally consistent SDRTV-to-HDRTV reconstruction, we construct a new dynamic HDRTV video dataset called TempHDRTV. It contains 74 video clips with a total of 5,535 frames. This dataset provides valuable

dynamic HDRTV content for training and evaluating video-based HDR reconstruction methods.

In summary, the main contributions of this work are as follows:

- We propose RB-HDRDiff, a diffusion-based HDRTV reconstruction framework that leverages a non-compressive latent space to effectively reduce banding artifacts, overcoming limitations of encoder-decoder ITM methods.
- We introduce a HDR Temporal Consistency Refiner, which incorporates temporal attention to enhance temporal coherence in HDRTV reconstruction.
- We construct the TempHDRTV video dataset consisting of 74 video clips to support research on temporally consistent SDRTV-to-HDRTV reconstruction.
- Extensive experiments demonstrate that RB-HDRDiff achieves superior visual fidelity and objective performance compared with state-of-the-art SDRTV-to-HDRTV methods.

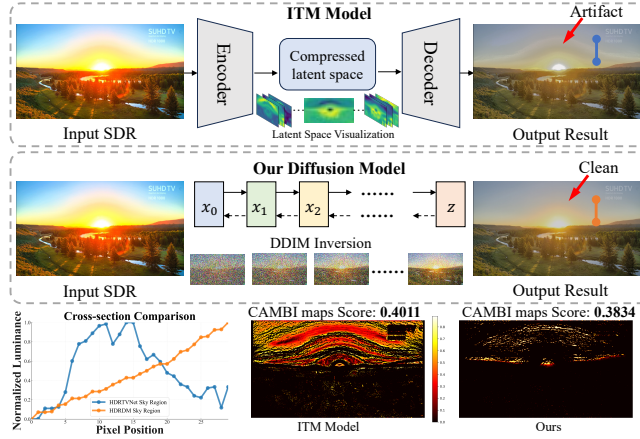


Fig. 2: Top: The ITM Model pipeline (Encoder-Decoder) processes introduces visual artifacts (marked by the blue arrow). Latent space visualization is shown for the compressed representation. Middle: Our Diffusion model pipeline leverages its non-compressive latent space to generate a clean HDR output (artifact-free, marked by the orange arrow). Lower Left: Cross-section comparison of normalized luminance between ITM Model (blue) and Ours (orange) across pixel positions. Lower Right: CAMBI maps with scores: higher for more banding artifacts demonstrating reduced perceptual distortion in our method.

2 Related Works

2.1 HDRTV Reconstruction

Numerous learning-based HDR methods have been developed, but their application scopes vary widely [5]. In this work, HDRTV refers to the display-referred

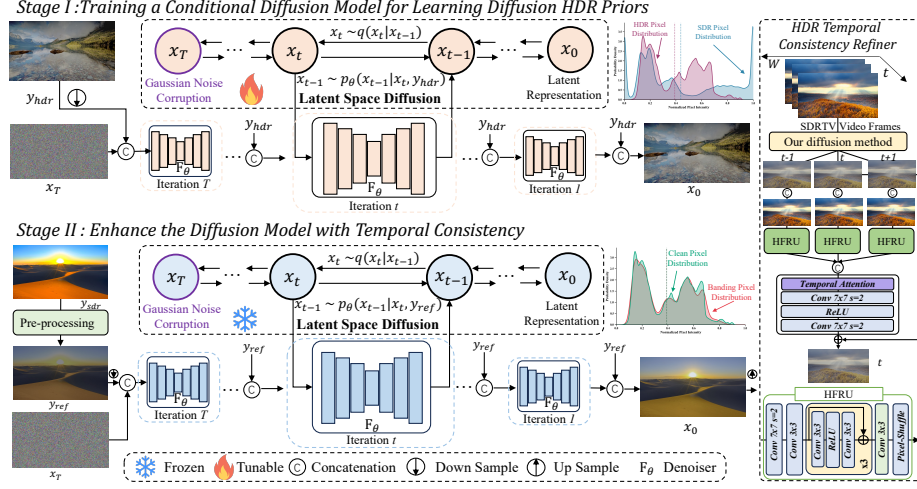


Fig. 3: Overview of our proposed pipeline. Our method consists of two stages. In the stage I, the diffusion model learns clean HDR distribution through learning from a large amount of high-quality data. Stage II using SDR-in-HDR pre-processing, the reference frames as conditions to guide the diffusion model. The reconstructed HDR frames, together with the corresponding SDR frames, are then fed into the HDR Temporal Consistency Refiner with HFRU modules to ensure temporal coherence.

WCG-HDR format using PQ/HLG EOTFs [10]. Early SDRTV-to-HDRTV approaches mainly rely on CNN-based encoder-decoder frameworks [14, 13], which enhance contrast and recover details through multi-level decomposition or task-specific modulation. Representative examples include JSI-GAN [14], Deep SR-ITM [13]. Multi-stage color mapping pipelines [1], and imaging-inspired architectures such as KUNet [29] and HycondITM [23]. Recent methods further explore luminance-range decomposition [5], frequency-aware modulation [32], and luminance-chrominance separation [8]. Despite these advances, most approaches still adopt heavily compressed encoder-decoder designs, which limit representational capacity and often lead to detail loss and banding artifacts. Furthermore, these methods process video frames independently without cross-frame interaction, which often leads to temporal inconsistencies in reconstructed HDR videos.

2.2 Diffusion Model for Image Reconstruction

Diffusion models have recently shown strong capability in image restoration. Fei et al. [4] introduced GDP, which achieves high-fidelity reconstruction across diverse tasks using a single unconditional DDPM pretrained on ImageNet. DiffIR [31] further learns a compact restoration prior via diffusion. Chung et al. [2] demonstrate that initializing sampling from a forward-diffused latent rather than pure noise can substantially reduce steps and improve generation quality. Other works [40] accelerate diffusion by formulating more accurate ODE

solvers. DTPM [34], a two-stage diffusion-based image restoration model that learns texture priors through large-scale pre-training and applies conditional guidance to reduce diffusion randomness and improve structural and perceptual quality. LEDiff [28] leverages pretrained latent diffusion models to fuse exposures in latent space and fine-tunes on small HDR datasets to recover extreme-range details. Building on Latent Diffusion [21], our method exploits the perceptual equivalence of latent and image spaces, enabling reconstruction in a compact yet semantically faithful representation while minimizing information loss.

2.3 Debanding

Traditional banding removal methods rely on false-contour detection followed by smoothing filters such as Gaussian [3] or averaging [27], but these hand-crafted approaches require careful parameter tuning. Recent CNN-based methods include blind decontouring via mask prediction [39] and GAN-based suppression [42]. However, these methods are not tailored for HDRTV generated by ITM algorithms. Zhou et al. [41] propose a post-processing CNN for ITM outputs, yet its encoder-decoder design compresses visual information into low-dimensional latent spaces, increasing the risk of banding. In contrast, our diffusion-based framework leverages perceptual equivalence between latent and image spaces to intrinsically reduce banding during reconstruction, while explicitly modeling inter-frame temporal coherence for improved temporal consistency.

3 Proposed Method

3.1 Model Overview

As shown Figure 3. Our RB-HDRDiff framework comprises two stages: In Stage 1, we train a conditional diffusion model with non-compress latent feature constraints using high-quality HDRTV images. In Stage 2, an HDR Temporal Consistency Refiner enhances video frames by refining and aggregating neighboring HDR features through temporal attention layer, improving temporal coherence while preserving high-fidelity HDR reconstruction.

3.2 Stage I: Learning Banding Reduction Diffusion HDRTV Priors

In diffusion models, the latent space is designed to closely align with the perceptual image space [21], which effectively mitigates banding artifacts and facilitates the generation of high-quality images. Motivated by this property, we integrate HDRTV-specific priors into the diffusion framework, enabling the exploitation of a large corpus of high-quality HDR images as prior knowledge [34]. This integration substantially enhances the model’s generalization ability while simultaneously improving sampling efficiency. During Stage I training, we employ high-quality HDRTV images from the widely recognized HDRTV1K [1] and HDRTV4K [5] datasets, together with our TempHDRTV dataset, resulting in a

total of 10,248 training images. This stage allows the model to learn the underlying distribution of clean, banding-free HDRTV content. Our framework is built upon conditional denoising diffusion probabilistic models [7, 24, 22], which comprise a forward diffusion process and a reverse denoising process, as illustrated in Figure 3. In the diffusion process, the HDRTV frame is initially downsampled to 128×256 serves as the initial state of the diffusion process. The diffusion process can be described as:

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}) \quad (1)$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t|\sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

where $\mathcal{N}$ denotes the gaussian distribution; $t = 1, \dots, T$ determine the diffusion steps; hyper-parameters $\alpha_{1:T}$ are subject to $0 < \alpha_t < 1$, and determine the variance of the noise added at each iteration. We can characterize the distribution of x_t given x_0 by marginalizing out the intermediate steps as:

$$q(x_t|x_0) = \mathcal{N}(x_t|\sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I) \quad (2)$$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

We accelerate the iterative procedure over multiple steps [24]. The reverse process can be described as:

$$x_{prev} = \sqrt{\bar{\alpha}_{prev}} \left(\frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_t}{\bar{\alpha}_t} \right) + \sqrt{1 - \bar{\alpha}_{prev} - \sigma_t^2 \epsilon_t + \sigma_t^2 \epsilon} \quad (3)$$

$$\sigma_t^2 = \frac{(1 - \bar{\alpha}_{t-1})(1 - \alpha_t)}{1 - \bar{\alpha}_t}$$

where ϵ_t is the prediction of the denoising model, ϵ is pure gaussian noise, and x_{prev} does not have to be exactly one step ahead of x_t .

Inspired by [2], selecting a more suitable starting point for the reverse process can improve the speed and quality of diffusion model generation. In the diffusion process, x_t is derived by adding noise to HDRTV frames. Hence, if we select initialization samplings that closely resemble the x_t , the resulting diffusion model generation will also better approximate the HDRTV frames. Therefore, we downsampled the pre-inverse tone mapping results obtained in the first stage, added noise, and utilized them as initialization samples. The Denoising UNet is constructed using the UNet2DModel module provided by Diffusers [20].

3.3 Stage II: Adapting model with Temporal consistency

RB-HDRDiff leverages strong priors learned from large-scale HDR data. To preserve this knowledge and avoid catastrophic forgetting, the diffusion model is frozen during adaptation. In this stage, the frozen model is guided by an HDR-conditioned input obtained via a deterministic SDR-in-HDR pre-processing pipeline.

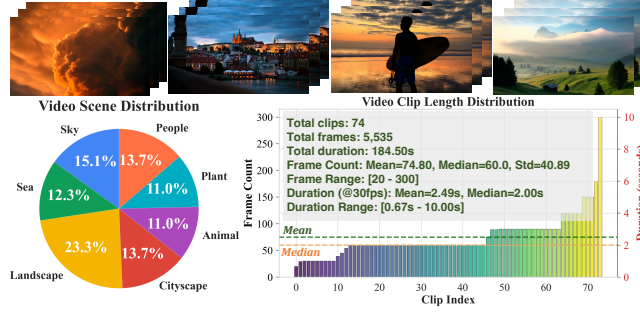


Fig. 4: Overview of the TempHDRTV video dataset, including clip length distribution, category proportions, and scene samples.

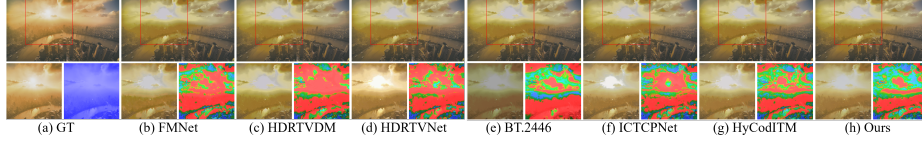


Fig. 5: Qualitative comparison between the Ground Truth HDR, and the results of other SOTA method. The error maps are shown below the results (color red indicate output result is different from the HDR).

Specifically, the input SDR frame (BT.709, $\gamma = 2.4$) is linearized, converted to CIE XYZ, mapped to linear BT.2020 RGB, and encoded using PQ (ST.2084) to produce an intermediate HDR reference y_{ref} . This pipeline performs a mathematically invertible, non-compressive domain transformation, avoiding projection into low-dimensional latent spaces. As a result, it preserves signal precision and introduces no additional quantization or false-contour artifacts, yielding a banding-free and structurally faithful HDR conditioning signal.

HDR Temporal Consistency Refiner (HTCR). After diffusion-based reconstruction, HTCR is applied to enhance temporal coherence. Given three consecutive reconstructed frames $\hat{I}_{t-1}, \hat{I}_t, \hat{I}_{t+1}$, HTCR predicts a temporally consistent estimate for the central frame $\hat{I}_t$. The module consists of three HDR Frame Refine Units (HFRUs), each processing one input frame independently. Within each HFRU, a low-resolution, banding-free HDR prior that captures the global luminance structure in the stage I is deterministically upsampled to full resolution via bicubic interpolation in the linear HDR domain. This process introduces neither quantization errors nor hallucinated details, thereby faithfully preserving the predicted global luminance distribution. The upsampled HDR prior is then concatenated with its corresponding SDR input, enabling high-frequency textures from the SDR signal to complement the HDR reconstruction. This fusion not only enriches spatial details but also contributes to improved temporal stability across frames.

HDR Frame Refine Units (HFRU). Each HFRU is a lightweight CNN with a large shallow receptive field. It begins with a 7×7 convolution (stride 2, 128 channels), followed by a 3×3 convolution reducing channels to 64. The back-

Table 1: Quantitative comparisons of HDRTV methods. The best and second best results are marked in **bold** and underlined.

HDRTV1K dataset										
Metric \ Method	Perceptual Uniform				Luma	HDR	Chroma	Conventional		Banding
	PSNR(Y) $\uparrow$	MS-SSIM $\uparrow$	VSI $\uparrow$	FSIM $\uparrow$	SR-SIM $\uparrow$	HDR-VDP3 $\uparrow$	ΔE_{ITP} $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	CAMBI $\downarrow$
BT2446-C	22.9130	0.9798	0.9938	0.9820	0.9867	7.8602	50.8072	23.19	0.9445	0.4693
JSI-GAN	36.9570	0.9866	0.9970	0.9900	0.9951	8.2618	11.7860	35.12	0.9609	0.4825
HyCondITM	39.4389	0.9889	<u>0.9982</u>	0.9930	0.9963	8.5262	9.0730	37.38	0.9733	0.4289
FMNet	40.0837	0.9892	0.9978	0.9926	0.9962	8.5564	8.0957	37.94	0.9747	0.4257
HDRTVNet	40.0150	0.9892	0.9980	0.9928	0.9967	<u>8.6121</u>	8.8924	37.62	0.9726	<u>0.4229</u>
HDRTVDM	39.3267	0.9888	0.9977	0.9929	0.9966	8.5516	10.0462	36.92	0.9717	0.4299
ICTCPNet	<u>40.5874</u>	<u>0.9895</u>	0.9981	<u>0.9938</u>	<u>0.9971</u>	8.5701	7.7877	<u>38.49</u>	0.9743	0.4392
KUNet	39.8651	0.9887	0.9976	0.9902	0.9941	8.3333	11.9283	37.78	<u>0.9748</u>	0.4432
Ours	43.4210	0.9989	0.9986	0.9989	0.9988	8.7409	<u>7.8092</u>	38.83	0.9824	0.3994
HDRTV4K dataset										
BT2446-C	22.4087	0.9809	0.9904	0.9762	0.9808	7.6819	60.7593	21.98	0.9253	0.4845
FMNet	<u>39.4762</u>	0.9886	<u>0.9940</u>	0.9848	<u>0.9895</u>	<u>8.3607</u>	<u>12.5348</u>	<u>36.30</u>	<u>0.9548</u>	<u>0.4434</u>
HDRTVNet	38.5898	0.9891	0.9939	<u>0.9860</u>	0.9899	8.3578	14.5282	35.65	0.9537	0.4726
HDRTVDM	33.0843	0.9431	0.9794	0.9620	0.9698	6.5106	44.5922	31.36	0.9083	0.4569
ICTCPNet	39.0786	<u>0.9832</u>	0.9937	0.9853	0.9890	8.2775	13.5711	35.73	0.9524	0.4734
Ours	40.3661	0.9912	0.9959	0.9878	0.9917	8.5999	8.9028	37.37	0.9626	0.4304

bone contains three ResBlocks without batch normalization to preserve absolute luminance. A pixel-shuffle layer restores spatial resolution, and a final 3×3 convolution produces the HDR output:

$$\begin{aligned}
 \text{ResBlock}(I) &= I + \text{Conv}_{k3s1c64} \circ \text{ReLU} \circ \text{Conv}_{k3s1c64}(I) \\
 \text{HFRU}(I) &= \text{Conv}_{k3s1c64} \circ \text{PixelShuffle} \circ \text{ResBlock}^3 \circ \\
 &\quad \text{Conv}_{k3s1c64} \circ \text{Conv}_{k7s2c128}(I)
 \end{aligned} \tag{4}$$

The triplet outputs are fed into a temporal attention layer [43], which adaptively aggregates temporal information to produce the HDR frame at time t . This design enhances temporal consistency while preserving reconstruction quality. Unlike flow-based methods [33], HTCR is trained end-to-end on spatio-temporal patches without precomputed optical flow, avoiding artifacts from inaccurate motion estimation in challenging scenarios such as occlusions [26]. By implicitly modeling motion, HTCR effectively exploits temporal redundancy to generate coherent HDR results.

3.4 TempHDRTV Video Dataset

We construct an HDRTV video dataset by collecting paired 4K resolution SDR-HDR video content from YouTube and segmenting them into scene-consistent short clips. As shown in Figure 4, the dataset covers a diverse range of scenes, including animals, humans, sky, and ocean environments, ensuring rich content variation in both spatial details and motion patterns. Each clip has a duration ranging from 1 to 10 seconds and is sampled at 29.97 frames per second. All HDR videos are encoded in BT.2020 gamut [9] with the PQ-EOTF and include complete HDR metadata, making the dataset well-suited for SDRTV-to-HDRTV

Table 2: Performance comparisons in terms of temporal consistency in our TempHDRTV dataset.

Metric	Perceptual Uniform				Luma	HDR	Chroma	Temporal		
	PSNR(Y) $\uparrow$	MS-SSIM $\uparrow$	VSI $\uparrow$	FSIM $\uparrow$	SR-SIM $\uparrow$	HDR-VDP3 $\uparrow$	ΔE_{JTF} $\downarrow$	FovVideoVDP $\uparrow$	ColorVideoVDP $\uparrow$	tPSNR $\uparrow$
FMNet	42.9753	0.9935	0.9981	0.9907	0.9951	8.1294	5.6356	9.3475	9.6982	41.7780
HDRTVDM	40.3294	0.9930	0.9978	0.9920	0.9927	7.9394	6.5796	9.3016	9.6473	41.1436
HDRTVNet	<u>43.1512</u>	0.9930	0.9983	<u>0.9929</u>	0.9934	8.0560	5.5843	9.3984	9.6727	<u>42.5006</u>
ICTCPNet	41.7454	<u>0.9938</u>	<u>0.9989</u>	0.9926	<u>0.9968</u>	<u>8.1477</u>	<u>5.4133</u>	<u>9.4348</u>	<u>9.7167</u>	42.1151
Ours	43.7454	0.9942	0.9992	0.9934	0.9988	8.2369	5.1347	9.4488	9.8132	42.7327

reconstruction and temporal consistency evaluation. In total, the dataset contains 74 video clips with 5,535 frames, corresponding to an overall duration of 184.5 seconds.

4 Experiments and Analysis

Dataset. We evaluate our method on the HDRTV1K, HDRTV4K and our TempHDRTV datasets. HDRTV1K contains 22 HDR10 video sequences with paired SDR counterparts from YouTube. All HDR frames follow PQ-EOTF, use the BT.2020 gamut, and have a resolution of 2160×3840 , yielding 1235 training and 117 test frames. For HDRTV4K, following [1, 5, 32], we adopt the YouTube tone-mapping operator to generate training pairs and use the official split from [5], which includes 3,478 training images and 400 test images. Our HDRTV video dataset contains 74 video clips, of which 60 are used for training and 14 for testing.

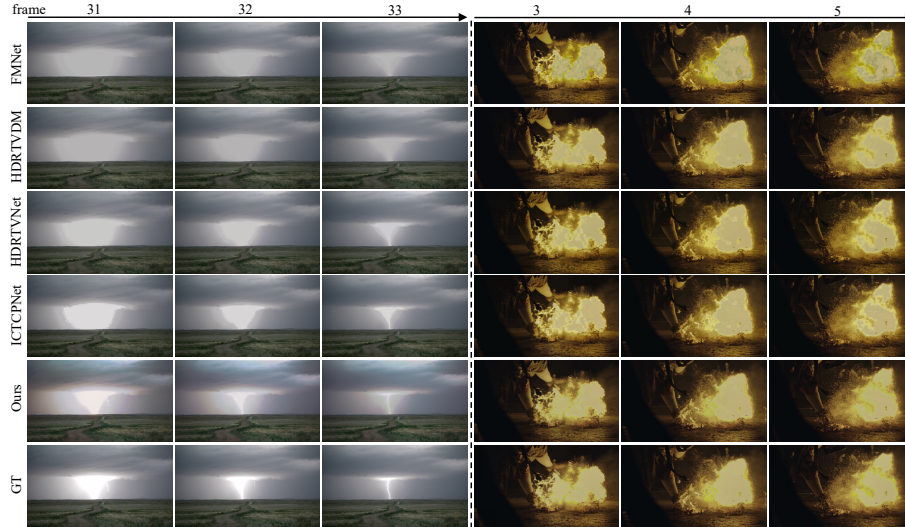


Fig. 6: Qualitative results in continuous video frames.

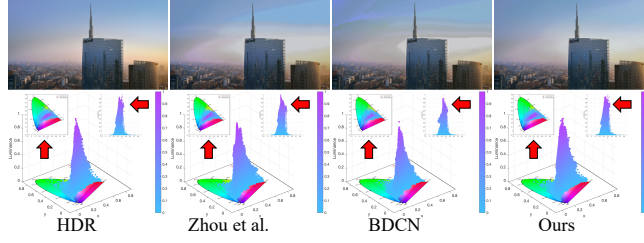


Fig. 7: Comparison of Debanding method. Pixel distribution (CIE xyY space): right subfig illustrations show chromaticity (xy) distribution, left subfig illustrations show luminance (Y).

Table 3: Performance comparisons of debanding methods in HDRTV1K

Methods	PSNR $\uparrow$	ΔE_{ITP} $\downarrow$	CAMBI $\downarrow$	HDR-VDP3 $\uparrow$
AdaDeband	37.76	9.7712	0.3594	7.4648
deepDeband	37.79	9.7793	0.3532	7.5539
BDCN	37.80	9.7865	0.3413	7.8644
Zhou et al.	<u>37.82</u>	<u>9.2826</u>	<u>0.3349</u>	<u>7.4623</u>
Ours	38.83	7.8092	0.3328	8.7409

Table 4: The ablation studies on HDRTV1K dataset

Methods	PSNR $\uparrow$	ΔE_{ITP} $\downarrow$	CAMBI $\downarrow$	HDR-VDP3 $\uparrow$
w/o Stage I Training	27.91	11.7812	0.5547	7.5235
w/o SDR	36.34	9.6542	0.4172	8.2539
w/o Temporal attn.	37.24	9.9901	0.4264	8.6367
w/o HTCR	37.93	9.8901	0.4013	8.1784
Ours	38.83	7.8092	0.3994	8.7409

Evaluation Metrics. Since HDR intensities are perceptually nonlinear, conventional metrics such as PSNR and SSIM are evaluated in a perceptually uniform (PU) space [16]. Following [15], HDR images are first encoded using a PU encoder, after which we report PSNR, PSNR(Y), SSIM, MS-SSIM [30], VSI [36], and FSIM [37]. In addition, we include HDR-VDP3 [18] and the chromaticity error defined in ITU-R BT.2124 [11]. To assess banding artifacts, we adopt the contrast-aware multiscale banding index (CAMBI) [25]. For completeness, conventional PSNR and SSIM, as well as SR-SIM [35], are also reported. For temporal consistency in videos, we employ FovVideoVDP [17], ColorVideoVDP [19], and tPSNR.

Compared Methods. We compare RB-HDRDiff with state-of-the-art SDRTV-to-HDRTV approaches, including KUNet [29], FMNet [32], JSI-GAN [14], HyCondITM [23], HDRTVNet [1], HDRTVDM [5], ICTCPNet [8], and the non-learning baseline BT.2446-C [12]. We also compare with some state-of-the-art debanding techniques, AdaDeband [27], deepDeband [42], BDCN [39] and Zhou et al [41]. For the handcrafted methods [27], we apply them directly to the ITM results since they do not have learnable parameters. As for deep learning-based approaches [42, 39, 41], with learnable parameters, we retained them using the HDRTV1K dataset.

Implementation Details. The diffusion model (stage one) is trained on 128×256 video frames obtained by bicubic downsampling from 4K. Stage two uses full-resolution images. Training is performed on a single RTX 4090 GPU using PyTorch, with batch sizes of 4. Stages 1-2 run for 300k and 100k iterations respectively, taking a total of 30 hours. We use L1 loss throughout, an initial learning rate of $2e-4$, stepLR (halved every 100k iterations) for stages 2, and cosine annealing for stage 1.

Quantitative Comparison. The quantitative results on HDRTV1K and HDRTV4K are summarized in Table 1. Our method achieves state-of-the-art performance across all HDRTV metrics, including HDR-VDP3, ΔE_{ITP} , PSNR, SSIM, MS-SSIM, VSI, FSIM, SR-SIM, and CAMBI, demonstrating superior perceptual and structural quality. Temporal evaluation on TempHDRTV (Table 2) further shows that our approach consistently outperforms existing methods in FovVideoVDP, ColorVideoVDP, and tPSNR, confirming its strong temporal stability and perceptual coherence. Moreover, as reported in Table 3, our method surpasses dedicated debanding approaches on HDRTV1K, achieving better HDR-VDP3, ΔE_{ITP} , and CAMBI scores, which highlights its effectiveness in reducing banding artifacts.

Quantitative Comparison. Figures 5 and 6 show qualitative comparisons of HDR image and video reconstruction. Our method produces clean results that closely match the ground truth in hue and luminance, while other methods suffer from noticeable banding artifacts. For dynamic scenes, competing approaches exhibit temporal banding artifacts. In contrast, our method maintains stable highlights, sharp structures, and smooth temporal transitions, yielding results that are visually consistent with the ground truth. Figure 7 visualizes debanding performance in the CIE xyY space. Baseline and competing methods exhibit discontinuous luminance and chromaticity distributions due to residual banding. Our method achieves smooth, continuous distributions in both channels, effectively reducing banding while preserving fine details and perceptual naturalness. More results are provided in the Supplementary Material.

4.1 Ablation Study

We conduct ablation studies on the HDRTV1K dataset to analyze the contribution of each key component. Quantitative results are reported in Table 4.

w/o Stage I Training. Removing the initial training stage leads to a substantial performance degradation across all metrics. PSNR drops sharply to 27.91, while ΔE_{ITP} increases to 11.7812, indicating poor perceptual fidelity. In addition, CAMBI and HDR-VDP3 deteriorate significantly, demonstrating that Stage I training is critical for stabilizing HDR reconstruction and providing a reliable initialization for subsequent refinement.

w/o SDR. Excluding the SDR input and directly reconstructing HDR content results in reduced perceptual quality. Although PSNR remains relatively high (36.34), both ΔE_{ITP} and CAMBI increase noticeably, and HDR-VDP3 decreases to 8.2539. This confirms that the SDR guidance is essential for preserving color consistency and perceptual realism during HDRTV reconstruction.

w/o Temporal Attention. Removing the temporal attention mechanism causes a decline in perceptual and temporal quality. While PSNR remains comparable, ΔE_{ITP} and CAMBI increase, and HDR-VDP3 drops from 8.7409 to 8.8367. These results indicate that temporal attention effectively aggregates inter-frame information to enhance perceptual coherence and reduce temporal artifacts.

w/o HTCR. Replacing the proposed HTCR module results in degraded perceptual performance, particularly in color accuracy and visual quality. Compared

with the full model, ΔE_{ITP} increases to 9.8901 and HDR-VDP3 decreases to 8.1784, highlighting the importance of HTCR in modeling spatio-temporal dependencies and maintaining perceptually consistent HDR outputs. More experiment detail in the Supplementary Material.

5 Conclusion

We present RB-HDRDiff, a diffusion-based framework for SDRTV-to-HDRTV reconstruction that reduces banding and enhances perceptual quality and temporal consistency. Leveraging the perceptually aligned latent space of diffusion models, it overcomes information loss and false contours typical of encoder-decoder ITMs. A two-stage training strategy with the HDR Temporal Consistency Refiner enables stable, coherent HDR video reconstruction without explicit motion estimation. We also introduce the TempHDRTV dataset to support research in temporally consistent HDRTV. Experiments show RB-HDRDiff outperforms state-of-the-art methods in visual fidelity and temporal stability.

References

1. Chen, X., Zhang, Z., Ren, J.S., Tian, L., Qiao, Y., Dong, C.: A new journey from sdrtv to hdrtv. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 4500–4509 (2021)
2. Chung, H., Sim, B., Ye, J.C.: Come-closer-diffuse-faster: Accelerating conditional diffusion models for inverse problems through stochastic contraction. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 12413–12422 (2022)
3. Daly, S.J., Feng, X.: Decontouring: Prevention and removal of false contour artifacts. In: Human Vision and Electronic Imaging IX. vol. 5292, pp. 130–149. SPIE (2004)
4. Fei, B., Lyu, Z., Pan, L., Zhang, J., Yang, W., Luo, T., Zhang, B., Dai, B.: Generative diffusion prior for unified image restoration and enhancement. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 9935–9946 (2023)
5. Guo, C., Fan, L., Xue, Z., Jiang, X.: Learning a practical sdr-to-hdrtv up-conversion using new dataset and degradation models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 22231–22241 (2023)
6. Guo, Z., Shao, M., Li, S.: Image-to-image translation using an offset-based multi-scale codes gan encoder. *The Visual Computer* **40**(2), 699–715 (2024)
7. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. *Advances in neural information processing systems* **33**, 6840–6851 (2020)
8. Huang, P., Cao, G., Zhou, F., Qiu, G.: Video inverse tone mapping network with luma and chroma mapping. In: Proceedings of the 31st ACM International Conference on Multimedia. pp. 1383–1391 (2023)
9. ITU-R: Parameter values for ultra-high definition television systems for production and international programme exchange., ITU-R Rec, BT.2020-2, Tech. Rep. (2015)
10. ITU-R: Image parameter values for high dynamic range television for use in production and international programme exchange. ITU-R Rec, BT.2100-2, Tech. Rep. (2018)

11. ITU-R: Objective metric for the assessment of the potential visibility of colour differences in television. ITU-R Rec, BT.2124-0, Tech. Rep. (2019)
12. ITU-R: Methods for conversion of high dynamic range content to standard dynamic range content and vice-versa. ITU-R Rec, BT.2446-1, Tech. Rep. (2021)
13. Kim, S.Y., Oh, J., Kim, M.: Deep sr-itm: Joint learning of super-resolution and inverse tone-mapping for 4k uhd hdr applications. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 3116–3125 (2019)
14. Kim, S.Y., Oh, J., Kim, M.: Jsi-gan: Gan-based joint super-resolution and inverse tone-mapping with pixel-wise task-specific filters for uhd hdr video. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 34, pp. 11287–11295 (2020)
15. Liu, P., Li, J., Wang, L., Zha, Z.J., Xiong, Z.: Mlp embedded inverse tone mapping. In: Proceedings of the 32nd ACM International Conference on Multimedia (MM '24). p. 9 pages. ACM, New York, NY, USA (October 28–November 1 2024). <https://doi.org/https://doi.org/10.1145/3664647.3680937>
16. Mantiuk, R.K., Azimi, M.: Pu21: A novel perceptually uniform encoding for adapting existing quality metrics for hdr. In: 2021 Picture Coding Symposium (PCS). pp. 1–5 (2021). <https://doi.org/10.1109/PCS50896.2021.9477471>
17. Mantiuk, R.K., Denes, G., Chapiro, A., Kaplanyan, A., Rufo, G., Bachy, R., Lian, T., Patney, A.: Fovvideovdp: a visible difference predictor for wide field-of-view video. ACM Trans. Graph. **40**(4) (Jul 2021). <https://doi.org/10.1145/3450626.3459831>, <https://doi.org/10.1145/3450626.3459831>
18. Mantiuk, R.K., Hammou, D., Hanji, P.: Hdr-vdp-3: A multi-metric for predicting image differences, quality and contrast distortions in high dynamic range and regular content. arXiv preprint arXiv:2304.13625 (2023)
19. Mantiuk, R.K., Hanji, P., Ashraf, M., Asano, Y., Chapiro, A.: Col-orvideovdp: A visual difference predictor for image, video and display distortions. ACM Trans. Graph. **43**(4) (Jul 2024). <https://doi.org/10.1145/3658144>, <https://doi.org/10.1145/3658144>
20. von Platen, P., Patil, S., Lozhkov, A., Cuenca, P., Lambert, N., Rasul, K., Davaadorj, M., Nair, D., Paul, S., Berman, W., Xu, Y., Liu, S., Wolf, T.: Diffusers: State-of-the-art diffusion models. <https://github.com/huggingface/diffusers> (2022)
21. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 10684–10695 (2022)
22. Saharia, C., Ho, J., Chan, W., Salimans, T., Fleet, D.J., Norouzi, M.: Image super-resolution via iterative refinement. IEEE transactions on pattern analysis and machine intelligence **45**(4), 4713–4726 (2022)
23. Shao, T., Zhai, D., Jiang, J., Liu, X.: Hybrid conditional deep inverse tone mapping. In: Proceedings of the 30th ACM International Conference on Multimedia. pp. 1016–1024 (2022)
24. Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models. arXiv preprint arXiv:2010.02502 (2020)
25. Tandon, P., Afonso, M., Sole, J., Krasula, L.: Cambi: Contrast-aware multiscale banding index. In: 2021 Picture Coding Symposium (PCS). pp. 1–5. IEEE (2021)
26. Tassano, M., Delon, J., Veit, T.: Fastdvdnet: Towards real-time deep video denoising without flow estimation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 1354–1363 (2020)
27. Tu, Z., Lin, J., Wang, Y., Adsumilli, B., Bovik, A.C.: Adaptive debanding filter. IEEE Signal Processing Letters **27**, 1715–1719 (2020)

28. Wang, C., Xia, Z., Leimkühler, T., Myszkowski, K., Zhang, X.: Lediff: Latent exposure diffusion for HDR generation. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025. pp. 453–464. Computer Vision Foundation / IEEE (2025)
29. Wang, H., Ye, M., Zhu, X., Li, S., Zhu, C., Li, X.: Kunet: Imaging knowledge-inspired single hdr image reconstruction. In: The 31st International Joint Conference On Artificial Intelligence (IJCAI/ECAI 22) (2022)
30. Wang, Z., Simoncelli, E.P., Bovik, A.C.: Multiscale structural similarity for image quality assessment. In: The Thirty-Seventh Asilomar Conference on Signals, Systems & Computers, 2003. vol. 2, pp. 1398–1402. Ieee (2003)
31. Xia, B., Zhang, Y., Wang, S., Wang, Y., Wu, X., Tian, Y., Yang, W., Van Gool, L.: Diffir: Efficient diffusion model for image restoration. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 13095–13105 (2023)
32. Xu, G., Hou, Q., Zhang, L., Cheng, M.M.: Fmnet: Frequency-aware modulation network for sdr-to-hdr translation. In: Proceedings of the 30th ACM International Conference on Multimedia. pp. 6425–6435 (2022)
33. Xu, G., Wang, Y., Gu, J., Xue, T., Yang, X.: Hdrflow: Real-time hdr video reconstruction with large motions. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 24851–24860 (2024)
34. Ye, T., Chen, S., Chai, W., Xing, Z., Qin, J., Lin, G., Zhu, L.: Learning diffusion texture priors for image restoration. In: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2524–2534 (2024). <https://doi.org/10.1109/CVPR52733.2024.00244>
35. Zhang, L., Li, H.: Sr-sim: A fast and high performance iqa index based on spectral residual. In: 2012 19th IEEE international conference on image processing. pp. 1473–1476. IEEE (2012)
36. Zhang, L., Shen, Y., Li, H.: Vsi: A visual saliency-induced index for perceptual image quality assessment. IEEE Transactions on Image processing **23**(10), 4270–4281 (2014)
37. Zhang, L., Zhang, L., Mou, X., Zhang, D.: Fsim: A feature similarity index for image quality assessment. IEEE transactions on Image Processing **20**(8), 2378–2386 (2011)
38. Zhang, X., Wu, X.: Lvqac: Lattice vector quantization coupled with spatially adaptive companding for efficient learned image compression. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10239–10248 (2023)
39. Zhao, Y., Jia, W., Chen, Y., Wang, R.: Fast blind decontouring network. IEEE Transactions on Circuits and Systems for Video Technology **33**(2), 478–490 (2022)
40. Zheng, K., Lu, C., Chen, J., Zhu, J.: Dpm-solver-v3: Improved diffusion ode solver with empirical model statistics. Advances in Neural Information Processing Systems **36** (2024)
41. Zhou, F., Zheng, Z., Qiu, G.: Removing banding artifacts in hdr videos generated from inverse tone mapping. IEEE Transactions on Broadcasting **70**(2), 753–762 (2024)
42. Zhou, R., Athar, S., Wang, Z., Wang, Z.: Deep image debanding. In: 2022 IEEE International Conference on Image Processing (ICIP). pp. 1951–1955. IEEE (2022)
43. Zhou, S., Yang, P., Wang, J., Luo, Y., Loy, C.C.: Upscale-a-video: Temporal-consistent diffusion model for real-world video super-resolution. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 2535–2545 (2024)